Finding the Next Signal, Round 1 Solution Report

Team APEX, **Track C, Consumer FMCG and Retail**, ATMOS 2026, BITS Pilani Hyderabad Campus

1. What we are building, and why it should find something new

Most tools in this space summarise what a company has already said. We want our agent to do the opposite. It reads what management says in investor presentations, earnings calls and the news, and then checks it against what the numbers of the company actually show. The interesting stuff usually sits in the gap between the two.

Every idea the agent comes up with has to get through three things, a calculation done in code, a check against a second source, and a proper attempt to prove it wrong. If it cannot defend a finding, it drops it and says so. We would rather show not enough evidence than force a story.

2. Why we picked Track C, FMCG

FMCG companies give out a lot of useful detail every quarter, like volume growth, price and mix, how general trade, modern trade and quick commerce are doing, plus receivables and inventory. That means the tests we care about, price versus volume, channel shift and working capital versus sales, can be worked out from public documents. Also, most consumer names in the NIFTY 50 report in a similar format, so one pipeline can handle all of them. Nothing is hard coded for a particular company. The agent starts from just the name.

3. Tools we plan to use

LLM	A good tool using model like Claude or Gemini for planning, pulling facts out of documents and running the red team step. We can swap the model if needed.
Agent loop	A simple Python orchestrator or LangGraph. It plans, collects, extracts, calculates, forms a hypothesis, challenges it and writes the report.
Sources	Web search plus direct reads of NSE and BSE filings, company investor relations pages with results, presentations, annual reports and call transcripts, MoSPI and RBI releases and reputed news. Public data only.
Parsing and maths	PyMuPDF and pdfplumber for PDFs and pandas for calculations. **The LLM never does the maths. All numbers are computed in code.**
Evidence store	A JSON or SQLite table where every fact keeps its value, period, source link, page, quote and how trustworthy the source is.
Front end	Streamlit app with a PDF export.

4. How it goes from a company name to a finding

1. Resolve	We type in only the company name. The agent works out the ticker, the sector and 3 to 5 peers.
2. Collect	It pulls the last 8 to 12 quarters of results, presentations, annual reports, exchange filings, call transcripts, and the same for peers.
3. Extract	Everything goes into one standard table with revenue, volume growth, price and mix, margins, receivables, inventory and channel comments. If a number is not disclosed, it is marked missing, not guessed.
4. Pick angles	It runs a fixed set of tests, like price versus volume, channel mix, receivable and inventory days versus sales, margins versus input costs, and what management says versus the numbers. Each one is scored by how unusual it is compared with the past of the company and its peers. The top 2 or 3 go forward.
5. Hypothesise	It writes down what changed, roughly when it started and a possible reason, and builds a quarter by quarter timeline to back it up.
6. Challenge	A separate red team step tries to break the finding, as explained in section 6. If it breaks, the agent moves to the next angle.
7. Report	It writes up the finding with every claim linked to its source.

5. Making it work for any company in the track

The pipeline is the same for every company and the only input is the name. Companies word things differently, one says underlying volume growth and another just says UVG, so the extraction step maps these onto common fields and marks anything that is not disclosed. Before the finals we will run it on at least 5 FMCG names, including one or two outside the NIFTY 50 like Dabur or Marico, in case of a tie break. If something breaks, we fix the general code, not a special patch for that company.

6. How the agent checks its own work

Triangulation. A finding needs at least 3 independent sources, of at least 2 different types, like the filings of the company, peer or industry data, and outside reporting. Filings and regulator data count the most, news the

least, and news on its own is never enough. The same figure is checked across the presentation, the results and the annual report, and any mismatch is flagged.

Trying to disprove itself. This is a separate step whose only job is to kill the finding.

- **Peer check.** If competitors show the same pattern, it is a sector story, not a company one, and gets downgraded.
- **Other explanations.** Base effects, one time items, accounting changes, mergers or demergers, seasonality, and price hikes driven by input costs.
- **Counter search.** It specifically goes looking for evidence against its own claim.
- **Falsification condition.** A clear number tied to a future date, usually the next quarterly result, that would prove the finding wrong.
- **Evidence Strength.** Strong, Moderate or Weak, decided by fixed rules everyone can see, like how many sources, how long the pattern has lasted, what the peer check said and which counterarguments it survived.

A quick example from our early prototype on HUL. At first the numbers seemed to say profit was running well ahead of cash. Once the agent took out the one time gain from the ice cream demerger, the gap disappeared, so it dropped that idea on its own. That is exactly the behaviour we want.

7. What the output looks like

A research report that shows up on screen and can be downloaded as a PDF. It opens with the main finding in one line and a strength rating, then 3 to 5 points on why it matters, 2 or 3 charts, for example price versus volume by quarter or receivable days versus peers, a timeline, an evidence table where every row links to the source and page, the counterarguments, the falsification condition and the reasoning behind the strength rating. No buy or sell advice anywhere.

8. Plan till the finals

Now	A basic version already runs end to end on HUL, with data collection, red team checks, a Streamlit screen and a PDF report.
Before shortlist	Better data collection with transcript quotes and the price and volume split, automatic peers, and the full set of angle tests.
After shortlist	Test on 5 or more companies including some outside the NIFTY, and fix whatever breaks in the general code.
Day before finals	Speed and backup. Cache data, add timeouts, record a backup demo run, and practise the 10 minute talk and the questions round.
25 Oct	Finals, with a live run on the company the judges pick.